

MIMIC-III vs MIMIC-IV: Dataset Comparison Summary

Generated from: `mimic_analysis/compute_part1.py` (Part 1) · `mimic_analysis/build_patient_table2.py` (Part 2) **Pipeline:** `build_mimic3_wide.py` · `build_mimic4_wide.py`

Part 1: Wide-Table Overview (All ICU Stays)

Inclusion: All ICU stays; one row per stay. All statistics computed over `hr ≥ 0` rows only.

Definitions

Time axis. Both datasets extend to pre-ICU hours (negative `hr`):

	MIMIC-III	MIMIC-IV
Pre-ICU window	<code>hr = -12</code> to discharge	<code>hr = -24</code> to discharge
Reason for difference	Official <code>pivoted_lab.sql</code> uses a ±12 h fuzzy window; labs up to 12 h before admission map to <code>hr = 0</code> , so extending beyond -12 produces empty rows	<code>labevents</code> is queried by raw <code>charttime</code> with no restriction, enabling capture of labs up to 24 h before ICU admission

For `hr < 0`, ICU-sourced variables (vitals, GCS, vasopressors, urine output, ventilation, SOFA, SepsisLabel) are NULL by construction; only hospital laboratory values may be non-null.

SOFA score. Computed hourly for `hr ≥ 0`. Each of the six components (respiration, coagulation, liver, cardiovascular, CNS, renal) is scored from raw values at the current hour (no carry-forward), then the maximum over the preceding 24 hours is taken (`MAX(score) OVER ROWS BETWEEN 24 PRECEDING AND CURRENT ROW`). `sofa_24hours` = sum of the six 24 h-max component scores.

Sepsis label.

	MIMIC-III	MIMIC-IV
Suspected infection	Antibiotic + culture within [-24 h, +12 h]	Antibiotic + culture within [-48 h, +24 h]
Sepsis criterion	SOFA rise ≥ 2 from 24 h rolling minimum (<code>sofa_delta_24h ≥ 2</code>)	Absolute SOFA ≥ 2 (<code>sofa_24hours ≥ 2</code>)
Reference	Seymour et al. 2016 / MIMIC-III Sepsis Challenge	Singer et al. 2016 / <code>mimic-code sepsis3.sql</code>

`SepsisLabel = 1` from `onset_hr` onward within the stay; = 0 before onset; = NULL for `hr < 0`. A stay is classified as a **sepsis stay** if `SepsisLabel = 1` at any `hr ≥ 0`.

ICU LOS: $\text{MAX}(\text{hr})$ per stay (hours from admission to discharge). **Coverage rate:** proportion of stays with ≥ 1 non-null value for the variable at $\text{hr} = 0$.

Table 1. Dataset Characteristics

	MIMIC-III	MIMIC-IV
Scale		
Total ICU stays	52,361	94,437
Unique subjects	38,343	65,365
Total patient-hours ($\text{hr} \geq 0$)	5,133,428	8,219,121
ICU LOS, mean (h)	97.0	86.0
ICU LOS, median [IQR] (h)	49.0 [27.0, 98.0]	46.0 [24.0, 92.0]
LOS < 24 h, %	19.1%	23.2%
Hospital mortality, n (%)	6,407 (12.2%)	11,347 (12.0%)
Sepsis		
Sepsis stays, n (%)	16,026 (30.6%)	41,296 (43.7%)
Non-sepsis stays, n (%)	36,335 (69.4%)	53,141 (56.3%)
Sepsis subjects, n (%)	13,161 (34.3%)	31,911 (48.8%)
Sepsis onset, median [IQR] (h)	2 [1, 11]	1 [0, 3]
SOFA		
Max SOFA per stay, mean	4.5	4.8
Max SOFA per stay, median [IQR]	4 [2, 6]	4 [2, 7]
Max SOFA ≥ 2 , % of stays	80.6%	81.9%
Max SOFA ≥ 6 , % of stays	31.2%	34.6%
SOFA at sepsis onset, median [IQR]	2 [1, 4]	3 [2, 4]
SOFA Δ at sepsis onset (Ill only), median [IQR]	1 [0, 2]	—

Table 2. First ICU Unit by Volume

ICU Unit	MIMIC-III	MIMIC-IV
Medical Intensive Care Unit (MICU)	—	20,699 (21.9%)
MICU	20,644 (39.4%)	—
Medical/Surgical Intensive Care Unit (MICU/SICU)	—	15,445 (16.4%)
Cardiac Vascular Intensive Care Unit (CVICU)	—	14,765 (15.6%)

ICU Unit	MIMIC-III	MIMIC-IV
Surgical Intensive Care Unit (SICU)	—	13,007 (13.8%)
Coronary Care Unit (CCU)	—	10,773 (11.4%)
Trauma SICU (TSICU)	—	10,472 (11.1%)
CSRU	9,154 (17.5%)	—
SICU	8,724 (16.7%)	—
CCU	7,572 (14.5%)	—
TSICU	6,267 (12.0%)	—
Neuro Intermediate	—	5,775 (6.1%)
Neuro Surgical Intensive Care Unit (Neuro SICU)	—	1,751 (1.9%)

Table 3. Clinical Data Coverage at hr = 0 (% of stays with ≥ 1 non-null value)

Variable	MIMIC-III	MIMIC-IV
Heart rate	98.4%	88.4%
Systolic BP	96.9%	84.3%
Temperature	N/A (not in wide table)	62.5%
SpO ₂	96.2%	86.0%
Creatinine	18.8%	20.6%
Platelet	18.5%	21.5%
Bilirubin (total)	6.3%	6.8%
GCS total	54.2%	43.8%
PaO ₂	18.8%	16.2%
FiO ₂	10.4%	8.6%

† Temperature is not available as a standalone vital in the MIMIC-III wide table (arterial blood gas temperature only).

Table 4. Sepsis Onset Hour Distribution (Sepsis Stays Only)

Onset hour = first **hr** at which **SepsisLabel** = **1** (always ≥ 0 ; consistent across both datasets). Each sepsis stay contributes exactly one value.

Dataset	Sepsis Stays	Mean (hr)	Median (hr)	P25 (hr)	P75 (hr)	SD	Min (hr)	Max (hr)
MIMIC-III	16,026	19.23	2.0	1.0	11.0	50.97	0	1277
MIMIC-IV	41,296	6.63	1.0	0.0	3.0	50.2	0	8769

Key Observations

1. Sepsis rate: MIMIC-IV (43.7%) > MIMIC-III (30.6%)

The difference is driven by the **labelling criterion**, not a genuinely higher prevalence of sepsis.

MIMIC-III applies a delta criterion (SOFA rise ≥ 2 from a 24 h rolling minimum), which does not label patients who arrive critically ill and do not deteriorate further. As a result, the median SOFA at sepsis onset is **2 [1, 4]**, indicating that most flagged patients experienced a rise from a low baseline rather than an acute decompensation. MIMIC-IV applies an absolute criterion (SOFA ≥ 2 at time of suspected infection), capturing patients admitted with an already-elevated SOFA; their median onset SOFA is **3 [2, 4]**.

The clinical severity of the labelled cohorts is nearly identical across datasets (first-day SOFA median 5.0 [3.0, 7.0]; 30-day mortality 20.9% vs 20.3%), confirming that the additional labels in MIMIC-IV represent patients missed by the structural blind spot of the delta criterion — those who arrive critically ill but do not deteriorate further — rather than a milder case mix.

2. Vitals coverage: MIMIC-IV (~55%) > MIMIC-III (~41%) at hr = 0

The gap reflects a documentation system transition: MIMIC-III was recorded primarily through CareVue (manual nursing entry), whereas MIMIC-IV transitioned to MetaVision (automated bedside monitor downloads), which captures vitals at higher frequency and completeness.

3. SOFA means aligned: MIMIC-III (4.5) $\approx$ MIMIC-IV (4.8)

Prior to correction, the MIMIC-III mean max SOFA was 8.3. The root cause was a structural defect in the renal SOFA urine-output logic, described in detail below.

In-Depth Analysis: Systematic Bias in Early Renal SOFA Scoring

Core Conclusion

When extracting the Renal SOFA score for the first 48 hours of an ICU admission, there is a massive gap between MIMIC-III and MIMIC-IV. Tracing the underlying SQL confirms: this is not a local data extraction bug, but a **severe boundary truncation defect in MIMIC-III's official script when handling cumulative metrics (urine output)**. The defect was structurally resolved in MIMIC-IV through an observation-time-aware mechanism.

I. Empirical Evidence: Systematic False Positives in MIMIC-III

ICU Hours Elapsed	MIMIC-III (Score = 4)	MIMIC-IV (Score = 4)	Observation
hr = 0	5.4%	0.5%	IV reflects true baseline of severe renal failure.
hr = 6	44.6% (peak)	—	III false positives peak; nearly half of patients deemed anuric.
hr = 24	49.5%	5.2%	IV completes first full 24 h window; ratio is stable.
hr = 48	8.6% (sudden drop)	7.5%	III exits the flawed rolling window; drops back to normal.

The absolute driver: urine output (UO). In MIMIC-III, 84.1% of abnormally high renal scores (≥ 2) are driven by UO alone:

- Pure UO trigger: **62.9%** — creatinine is normal; the patient is penalized solely by the UO logic.
- Pure creatinine trigger: 19.9%.

II. Defect Origin: Flaw in MIMIC-III's Official Script

The root cause lies in `pivoted_sofa.sql`. A fixed row-based window is applied blindly, producing two compounding errors: **early boundary truncation** and **extreme-value propagation**.

For snapshot metrics (respiration, coagulation), taking the min or max over 24 hours is unaffected by how long the patient has been in the ICU. Urine output is a **cumulative metric** requiring `SUM` — truncated sums produce systematically wrong totals.

MIMIC-III official `pivoted_sofa.sql`:

```
-- Root cause: mechanical 24-row backward rolling window
SUM(UrineOutput) OVER (
  PARTITION BY icustay_id
  ORDER BY hr
  ROWS BETWEEN 23 PRECEDING AND CURRENT ROW -- sums whatever rows exist,
  regardless of actual time span
) AS uo_24hr
```

Error chain:

1. **Truncated comparison** — At hr = 6 the engine finds only 7 rows. Normal urine over 7 hours (~400 mL) is compared directly against the 24 h clinical threshold (< 500 mL = score 3).
2. **False-positive explosion** — Seven hours of urine can rarely meet a 24 h passing mark; healthy kidneys are mass-classified as severely oliguric in the early hours.
3. **Ghost propagation** — The final SOFA uses `MAX(sofa_renal) OVER 24 PRECEDING`. A spuriously generated score of 4 at hr = 7 propagates forward for 24 hours, exiting the window completely only at hr = 48 — explaining the abrupt drop above.

III. The Fix: Observation-Time-Aware Logic in MIMIC-IV

MIMIC-IV's `sofa.sql` introduces `uo_tm_24hr` (accumulated observation duration) and refuses to score until the window is sufficiently covered.

MIMIC-IV official `sofa.sql`:

```
MAX(
  CASE
    -- Refuse to evaluate if observation window < 22 h
    WHEN uo.uo_tm_24hr >= 22 AND uo.uo_tm_24hr <= 30
    -- Normalise to a standard 24 h equivalent
    THEN uo.urineoutput_24hr / uo.uo_tm_24hr * 24
  END
) AS uo_24hr
```

During the first ~22 hours, `uo_24hr` is forcibly NULL — no UO-driven score is possible. The scoring engine falls back entirely to admission serum creatinine, which is why MIMIC-IV shows a realistic 0.5% severe-failure rate at hr = 0. This logic was ported into the MIMIC-III pipeline (`build_mimic3_wide.py`, `step07_uo`), after which both datasets align: max SOFA mean 4.5 (III) vs 4.8 (IV).

Part 2: Patient-Level Cohort Summary

Inclusion: One row per patient, first ICU stay only.

Definitions:

- **SOFA:** first-day maximum SOFA score
- **SIRS:** first-day maximum SIRS score
- **Comorbidity index:** Elixhauser van Walraven score (MIMIC-III, official mimic-code ICD-9 implementation) / Charlson Comorbidity Index with age adjustment (MIMIC-IV, official mimic-code ICD-9 + ICD-10 implementation). *Scores are not directly comparable across datasets.*
- **Mechanical ventilation:** any invasive/non-invasive ventilation or tracheostomy during the first ICU stay
- **30-day mortality:** measured from first ICU admission time
- **Race/ethnicity:** grouped into 7 categories (White, Black, Hispanic, Asian, Native, Other, Unknown)

Table 5. MIMIC-III Patient Characteristics

Variable	All patients (N = 38,343 (100.0%))	Survivors (N = 33,979 (88.6%))	Non-survivors (N = 4,364 (11.4%))	P-value (Survival)	Sepsis (N = 10,990 (28.7%))	Non-sepsis (N = 27,353 (71.3%))	P-value (Sepsis)
Age (y) [Q1-Q3]	66.0 [52.0, 78.0]	65.0 [52.0, 77.0]	74.0 [61.0, 83.0]	<0.001	67.0 [54.0, 79.0]	65.0 [52.0, 77.0]	<0.001

Variable	All patients (N = 38,343 (100.0%))	Survivors (N = 33,979 (88.6%))	Non-survivors (N = 4,364 (11.4%))	P-value (Survival)	Sepsis (N = 10,990 (28.7%))	Non-sepsis (N = 27,353 (71.3%))	P-value (Sepsis)
Male, n (%)	21711 (56.6%)	19413 (57.1%)	2298 (52.7%)	<0.001	6428 (58.5%)	15283 (55.9%)	<0.001
BMI (kg/m ²), mean +/- SD	39.5 +/- 977.2	40.6 +/- 1034.0	30.7 +/- 110.3	0.170	48.5 +/- 1390.9	35.1 +/- 692.6	0.407
Race, n (%)				<0.001			<0.001
White	27370 (71.4%)	24391 (71.8%)	2979 (68.3%)		7969 (72.5%)	19401 (70.9%)	
Black	2937 (7.7%)	2687 (7.9%)	250 (5.7%)		873 (7.9%)	2064 (7.5%)	
Hispanic	1247 (3.3%)	1162 (3.4%)	85 (1.9%)		335 (3.0%)	912 (3.3%)	
Asian	903 (2.4%)	802 (2.4%)	101 (2.3%)		271 (2.5%)	632 (2.3%)	
Native	20 (0.1%)	19 (0.1%)	1 (0.0%)		5 (0.0%)	15 (0.1%)	
Other	1036 (2.7%)	931 (2.7%)	105 (2.4%)		329 (3.0%)	707 (2.6%)	
Unknown	4830 (12.6%)	3987 (11.7%)	843 (19.3%)		1208 (11.0%)	3622 (13.2%)	
Comorbidity index [Q1- Q3]	5.0 [0.0, 12.0]	5.0 [0.0, 11.0]	11.0 [5.0, 17.0]	<0.001	9.0 [4.0, 16.0]	5.0 [0.0, 10.0]	<0.001
SIRS first- day max [Q1-Q3]	3.0 [2.0, 3.0]	3.0 [2.0, 3.0]	3.0 [2.0, 4.0]	<0.001	3.0 [2.0, 4.0]	2.0 [2.0, 3.0]	<0.001
SOFA first- day [Q1-Q3]	3.0 [1.0, 5.0]	2.0 [1.0, 4.0]	5.0 [3.0, 8.0]	<0.001	5.0 [3.0, 7.0]	2.0 [1.0, 4.0]	<0.001
ICU length- of-stay (d) [Q1-Q3]	2.0 [1.0, 4.0]	2.0 [1.0, 4.0]	3.0 [1.0, 7.0]	<0.001	4.0 [2.0, 9.0]	2.0 [1.0, 3.0]	<0.001
Mechanical ventilation, n (%)	19473 (50.8%)	16263 (47.9%)	3210 (73.6%)	<0.001	7576 (68.9%)	11897 (43.5%)	<0.001

Variable	All patients (N = 38,343 (100.0%))	Survivors (N = 33,979 (88.6%))	Non-survivors (N = 4,364 (11.4%))	P-value (Survival)	Sepsis (N = 10,990 (28.7%))	Non-sepsis (N = 27,353 (71.3%))	P-value (Sepsis)
30-day mortality, n (%)	5204 (13.6%)	1074 (3.2%)	4130 (94.6%)	<0.001	2293 (20.9%)	2911 (10.6%)	<0.001
Hospital mortality, n (%)	4364 (11.4%)	0 (0.0%)	4364 (100.0%)	<0.001	2041 (18.6%)	2323 (8.5%)	<0.001

Table 6. MIMIC-IV Patient Characteristics

Variable	All patients (N = 65,365 (100.0%))	Survivors (N = 58,279 (89.2%))	Non-survivors (N = 7,086 (10.8%))	P-value (Survival)	Sepsis (N = 27,890 (42.7%))	Non-sepsis (N = 37,475 (57.3%))	P-value (Sepsis)
Age (y) [Q1-Q3]	66.0 [54.0, 78.0]	66.0 [54.0, 77.0]	73.0 [61.0, 83.0]	<0.001	67.0 [56.0, 78.0]	65.0 [53.0, 77.0]	<0.001
Male, n (%)	36719 (56.2%)	32877 (56.4%)	3842 (54.2%)	<0.001	16160 (57.9%)	20559 (54.9%)	<0.001
BMI (kg/m ²), mean +/- SD	29.2 +/- 7.8	29.3 +/- 7.7	29.1 +/- 7.9	0.320	29.7 +/- 8.1	28.7 +/- 7.4	<0.001
Race, n (%)				<0.001			<0.001
White	43093 (65.9%)	38990 (66.9%)	4103 (57.9%)		18257 (65.5%)	24836 (66.3%)	
Black	6016 (9.2%)	5470 (9.4%)	546 (7.7%)		2366 (8.5%)	3650 (9.7%)	
Hispanic	2351 (3.6%)	2177 (3.7%)	174 (2.5%)		952 (3.4%)	1399 (3.7%)	
Asian	1981 (3.0%)	1769 (3.0%)	212 (3.0%)		817 (2.9%)	1164 (3.1%)	
Native	127 (0.2%)	117 (0.2%)	10 (0.1%)		59 (0.2%)	68 (0.2%)	
Other	2416 (3.7%)	2183 (3.7%)	233 (3.3%)		1012 (3.6%)	1404 (3.7%)	

Variable	All patients (N = 65,365 (100.0%))	Survivors (N = 58,279 (89.2%))	Non-survivors (N = 7,086 (10.8%))	P-value (Survival)	Sepsis (N = 27,890 (42.7%))	Non-sepsis (N = 37,475 (57.3%))	P-value (Sepsis)
Unknown	9381 (14.4%)	7573 (13.0%)	1808 (25.5%)		4427 (15.9%)	4954 (13.2%)	
Comorbidity index [Q1-Q3]	4.0 [2.0, 7.0]	4.0 [2.0, 6.0]	6.0 [4.0, 8.0]	<0.001	5.0 [3.0, 7.0]	4.0 [2.0, 6.0]	<0.001
SIRS first-day max [Q1-Q3]	2.0 [1.0, 2.0]	2.0 [1.0, 2.0]	2.0 [2.0, 3.0]	<0.001	2.0 [2.0, 3.0]	2.0 [1.0, 2.0]	<0.001
SOFA first-day [Q1-Q3]	3.0 [2.0, 6.0]	3.0 [1.0, 5.0]	6.0 [4.0, 10.0]	<0.001	5.0 [3.0, 7.0]	2.0 [1.0, 4.0]	<0.001
ICU length-of-stay (d) [Q1-Q3]	1.9 [1.1, 3.7]	1.9 [1.1, 3.5]	2.7 [1.1, 6.4]	<0.001	2.9 [1.5, 6.3]	1.5 [0.9, 2.5]	<0.001
Mechanical ventilation, n (%)	28444 (43.5%)	23589 (40.5%)	4855 (68.5%)	<0.001	17521 (62.8%)	10923 (29.1%)	<0.001
30-day mortality, n (%)	8970 (13.7%)	2208 (3.8%)	6762 (95.4%)	<0.001	5672 (20.3%)	3298 (8.8%)	<0.001
Hospital mortality, n (%)	7086 (10.8%)	0 (0.0%)	7086 (100.0%)	<0.001	4710 (16.9%)	2376 (6.3%)	<0.001

[†] Elixhauser van Walraven score (MIMIC-III): 29-condition weighted index based on secondary ICD-9 diagnoses (Quan et al. 2005; van Walraven et al. 2009). Scores range from approximately −19 to +89; higher scores indicate greater predicted in-hospital mortality risk. Negative weights apply to certain conditions (e.g., drug abuse −7, obesity −4).

[‡] Charlson Comorbidity Index with age adjustment (MIMIC-IV): 17-condition index based on ICD-9 and ICD-10 diagnoses (Charlson et al. 1987; Quan et al. 2011). Age contributes 0–4 points (1 point per decade above age 50). Scores range from 0 to approximately 30; higher scores indicate greater comorbidity burden. The two comorbidity indices are not directly comparable.

Part 4: Subject-Level ICU Stay Patterns

Unit of analysis: one row per unique subject (first vs. subsequent stays). Sepsis label per stay = 1 if `SepsisLabel = 1` at any `hr >= 0` within that stay, else 0.

4.1 Distribution of ICU Stays per Subject

MIMIC-III: **38,343** unique subjects - 8,140 (21.2%) have more than one ICU stay.

MIMIC-IV: **65,365** unique subjects - 16,229 (24.8%) have more than one ICU stay.

All non-sepsis = all stays for this subject are labelled 0; *All sepsis* = all stays labelled 1; *Mixed* = at least one 0 and at least one 1 across stays.

MIMIC-III

ICU stays	Subjects, n (%)	Cumulative (%)	All non-sepsis	All sepsis	Mixed
1	30,203 (78.8%)	78.8%	21,880 (72.4%)	8,323 (27.6%)	0 (0.0%)
2	5,368 (14.0%)	92.8%	2,614 (48.7%)	706 (13.2%)	2,048 (38.2%)
3	1,511 (3.9%)	96.7%	485 (32.1%)	120 (7.9%)	906 (60.0%)
4	623 (1.6%)	98.3%	133 (21.3%)	34 (5.5%)	456 (73.2%)
5	281 (0.7%)	99.1%	44 (15.7%)	13 (4.6%)	224 (79.7%)
6	142 (0.4%)	99.4%	12 (8.5%)	1 (0.7%)	129 (90.8%)
7	76 (0.2%)	99.6%	10 (13.2%)	0 (0.0%)	66 (86.8%)
8	39 (0.1%)	99.7%	0 (0.0%)	0 (0.0%)	39 (100.0%)
9	25 (0.1%)	99.8%	1 (4.0%)	0 (0.0%)	24 (96.0%)
10	23 (0.1%)	99.9%	0 (0.0%)	0 (0.0%)	23 (100.0%)
>=11	52 (0.1%)	100.0%	3 (5.8%)	0 (0.0%)	49 (94.2%)

MIMIC-IV

ICU stays	Subjects, n (%)	Cumulative (%)	All non-sepsis	All sepsis	Mixed
1	49,136 (75.2%)	75.2%	28,829 (58.7%)	20,307 (41.3%)	0 (0.0%)
2	10,331 (15.8%)	91.0%	3,722 (36.0%)	2,351 (22.8%)	4,258 (41.2%)
3	3,204 (4.9%)	95.9%	648 (20.2%)	480 (15.0%)	2,076 (64.8%)
4	1,291 (2.0%)	97.9%	167 (12.9%)	131 (10.1%)	993 (76.9%)
5	578 (0.9%)	98.7%	40 (6.9%)	44 (7.6%)	494 (85.5%)
6	308 (0.5%)	99.2%	22 (7.1%)	13 (4.2%)	273 (88.6%)
7	158 (0.2%)	99.5%	5 (3.2%)	5 (3.2%)	148 (93.7%)
8	110 (0.2%)	99.6%	7 (6.4%)	5 (4.5%)	98 (89.1%)

ICU stays	Subjects, n (%)	Cumulative (%)	All non-sepsis	All sepsis	Mixed
9	56 (0.1%)	99.7%	3 (5.4%)	2 (3.6%)	51 (91.1%)
10	52 (0.1%)	99.8%	2 (3.8%)	1 (1.9%)	49 (94.2%)
>=11	141 (0.2%)	100.0%	9 (6.4%)	1 (0.7%)	131 (92.9%)

4.2 Sepsis Sequence Patterns

Each subject's ICU stays are ordered by admission time and collapsed to a binary sepsis label per stay (0 = no sepsis, 1 = sepsis), then concatenated into a sequence such as 0->1->0. Proportion within group = share among all subjects with the same number of ICU stays. The 1 stay, 2 stays, 3 stays, and 4 stays sections are mutually exclusive subject subgroups rather than nested subsets.

MIMIC-III: **294** unique sequences in total (30 with <=4 stays).

MIMIC-IV: **512** unique sequences in total (30 with <=4 stays).

MIMIC-III

Sequences with 1-4 ICU Stays (all combinations)

n stays	Sequence	Subjects, n	% within group
1	0	21,880	72.4%
1	1	8,323	27.6%
2	0->0	2,614	48.7%
2	0->1	1,067	19.9%
2	1->0	981	18.3%
2	1->1	706	13.2%
3	0->0->0	485	32.1%
3	0->0->1	218	14.4%
3	1->0->0	169	11.2%
3	0->1->0	159	10.5%
3	1->0->1	134	8.9%
3	0->1->1	124	8.2%
3	1->1->1	120	7.9%
3	1->1->0	102	6.8%
4	0->0->0->0	133	21.3%
4	0->0->0->1	62	10.0%

n stays	Sequence	Subjects, n	% within group
4	0->0->1->0	46	7.4%
4	1->0->0->0	45	7.2%
4	0->0->1->1	39	6.3%
4	0->1->0->0	39	6.3%
4	1->0->0->1	37	5.9%
4	1->1->1->1	34	5.5%
4	0->1->0->1	31	5.0%
4	1->0->1->1	27	4.3%
4	1->1->1->0	25	4.0%
4	0->1->1->0	24	3.9%
4	1->1->0->1	22	3.5%
4	0->1->1->1	20	3.2%
4	1->0->1->0	20	3.2%
4	1->1->0->0	19	3.0%

Sequences with >=5 ICU Stays (summary)

n stays	Subjects	Unique patterns	Most common sequence	Count
5	281	32	0->0->0->0->0	44
6	142	54	0->0->0->0->0->0	12
7	76	47	0->0->0->0->0->0->0	10
8	39	35	0->0->0->0->0->0->1->0	2
9	25	25	0->0->0->0->0->0->0->0->0	1
10	23	21	0->0->0->0->0->0->0->0->0->1	2
11	8	8	0->0->0->0->0->0->1->1->0->0->1	1
12	9	8	0->0->0->0->0->0->0->0->0->0->0	2
13	10	9	0->0->0->0->0->0->0->0->0->1->0->0->1	2
14	5	5	0->0->0->0->0->0->0->1->0->1->1->0->0->0	1
15	4	4	0->0->0->0->0->0->0->0->0->0->0->1->0->1->0->0	1
16	1	1	1->1->1->1->0->1->1->1->0->1->0->1->1->0->0->1	1

n stays	Subjects	Unique patterns	Most common sequence	Count
17	2	2	0->0->0->0->0->0->0->0->0->0->1->0->0->0->0->0	1
18	2	2	0->0->0->0->0->0->0->1->0->0->0->1->0->0->0->0->0->0	1
19	1	1	0->1->0->1->0->0->1->0->1->0->0->0->1->1->0->1->1->0->1	1
21	2	2	0->0->0->0->0->0->0->1->0->0->0->0->0->0->0->0->0->1->0->0->0	1
22	1	1	0->0->1->1->0->0->1->0->0->0->1->1->0->0->1->0->1->0->1->1->0->1	1
23	1	1	0->0->0->0->0->1->0->0->0->0->0->1->0->0->0->1->0->0->0->0->0->0	1
24	1	1	0->0->0->0->0->1->0->0->0->0->0->0->1->0->0->0->0->0->1->0->1->0->1->0->1	1
25	1	1	1->0->0->0->1->1->0->0->0->1->0->0->0->0->0->0->0->1->0->0->0->0->0->1	1
30	1	1	0->0	1
35	1	1	0->0->0->0->0->0->0->0->1->0->0->0->0->0->0->0->0->0->0->0->0->0->0->0	1
38	1	1	0->0->0->0->0->0->0->1->1->0->0->0->1->0->1->0->1->0->0->0->0->0->0->0->0->0->0->0->0->0->0->0	1
41	1	1	0->0->1->0->0->0->0->0->1->0->0->0->0->0->0->0->0->0->0->0->0->1->0->0->0->1->0->0->0->1->0->0->0->1->0->0->0->1->0->0->0->0	1

MIMIC-IV

Sequences with 1-4 ICU Stays (all combinations)

n stays	Sequence	Subjects, n	% within group
1	0	28,829	58.7%
1	1	20,307	41.3%
2	0->0	3,722	36.0%

n stays	Sequence	Subjects, n	% within group
2	1->1	2,351	22.8%
2	1->0	2,215	21.4%
2	0->1	2,043	19.8%
3	0->0->0	648	20.2%
3	1->1->1	480	15.0%
3	1->0->0	422	13.2%
3	0->0->1	361	11.3%
3	1->0->1	353	11.0%
3	1->1->0	340	10.6%
3	0->1->1	308	9.6%
3	0->1->0	292	9.1%
4	0->0->0->0	167	12.9%
4	1->1->1->1	131	10.1%
4	1->0->1->1	93	7.2%
4	0->0->0->1	90	7.0%
4	1->0->0->0	81	6.3%
4	1->0->0->1	81	6.3%
4	1->1->0->1	79	6.1%
4	0->1->0->0	75	5.8%
4	1->1->0->0	72	5.6%
4	0->0->1->0	67	5.2%
4	0->0->1->1	67	5.2%
4	0->1->1->1	67	5.2%
4	1->1->1->0	64	5.0%
4	0->1->0->1	57	4.4%
4	0->1->1->0	53	4.1%
4	1->0->1->0	47	3.6%

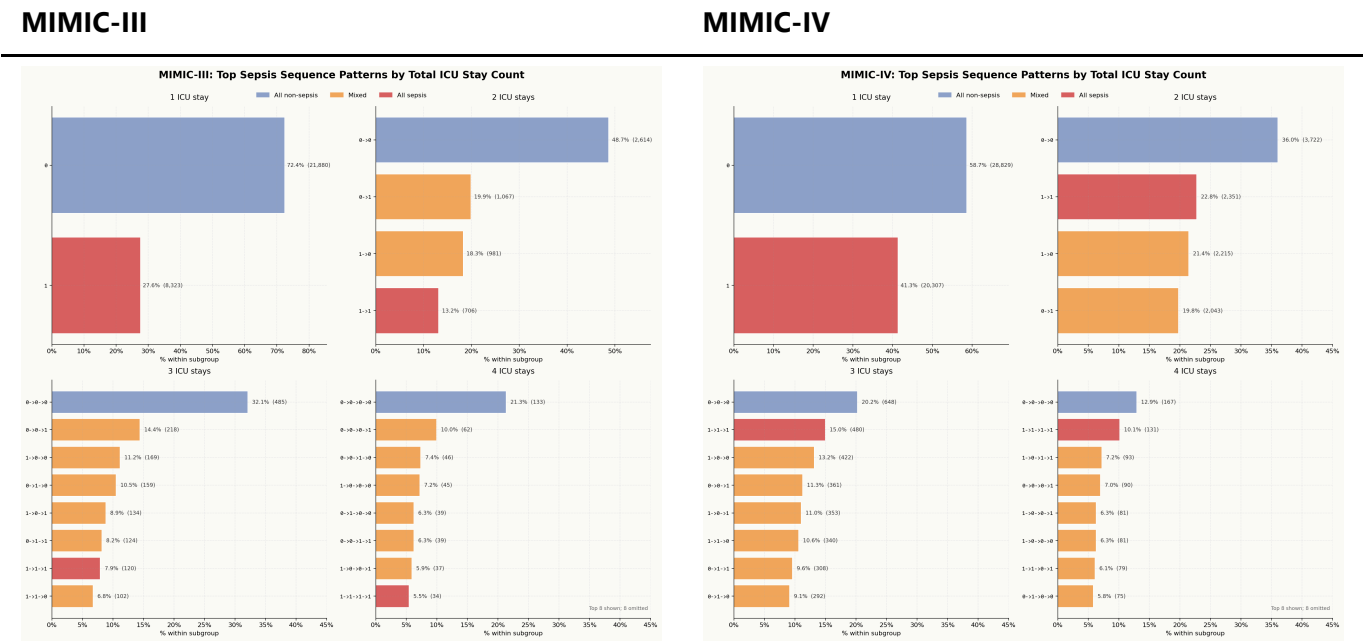
Sequences with >=5 ICU Stays (summary)

n stays	Subjects	Unique patterns	Most common sequence	Count
5	578	32	1->1->1->1->1	44
6	308	62	0->0->0->0->0->0	22
7	158	78	0->0->0->0->0->0->0	5
8	110	73	0->0->0->0->0->0->0->0	7
9	56	49	0->0->0->0->0->0->0->0->0	3
10	52	51	0->0->0->0->0->0->0->0->0->0	2
11	43	40	0->0->0->0->0->0->0->0->0->0->0	4
12	20	20	0->0->0->0->0->0->0->0->0->0->0->0	1
13	17	17	0->0->0->0->0->0->0->0->1->0->0->0->0->0	1
14	6	6	0->0->0->1->0->1->0->0->1->0->0->1->1->0	1
15	10	10	0->0->0->0->0->0->0->0->0->0->0->0->0->0	1
16	9	9	0->0->1->0->0->0->0->0->0->0->0->1->1->0->1->1->1	1
17	4	4	0->0->1->1->1->0->0->0->0->0->1->1->1->0->0->1-> >1	1
18	11	10	0->0->0->0->0->0->0->0->0->0->0->0->0->0->0->0->0->0->0	2
19	2	2	0->0->0->0->1->0->0->1->1->1->1->1->1->0->0->0->0->0->1	1
20	3	3	0->0->0->0->0->0->0->0->0->0->0->0->0->0->0->0->0->0->0	1
22	3	3	0->0->0->0->0->1->0->0->0->0->0->0->0->0->0->0->0->0->0->1	1
24	3	3	0->0->0->0->0->0->1->1->1->0->0->0->0->0->0->1->0->0->1->0->0->1	1
25	3	3	0->0->0->0->1->1->0->0->1->0->0->0->0->0->1->0->0->0->0->1->0->0->1->0->0->1->1	1
26	1	1	0->0->0->0->0->1->0	1
27	1	1	0->0->0->0->0->1->0->1->0->0->0->0->0->0->0->0->0->0->0->1->1->0->0->1->0->0->1->0->0->1	1
30	1	1	0->0->0->0->0->0->0->0->0->0->0->0->0->0->0->0->1->0->0->0->0->0->1->0->0->1->1->0->1->1->0->1	1

n stays	Subjects	Unique patterns	Most common sequence	Count
31	1	1	1->1->0->0->0->1->0->0->0->1->0->0->1->1->1->1-> >1->0->0->0->1->0->0->0->0->0->0->0->1->0	1
34	1	1	0->0->0->0->0->0->0->0->1->0->0->0->0->0-> >0->0->0->0->0->0->0->0->1->0->0->0->0->0-> >0->0	1
37	1	1	1->1->0->1->1->0->0->0->0->0->0->0->0->1-> >0->0->0->0->0->0->0->0->0->0->0->0->0->0-> >0->1->0->0->0	1
41	1	1	1->0->0->1->1->1->1->1->0->1->1->1->1->1->0-> >1->1->1->1->1->0->1->1->1->0->1->1->1->1->1->1-> >1->0->1->1->1->1->1->1->1	1

Visual Summary

The figure below groups subjects by their total number of ICU stays and shows the most common sequence patterns within each subgroup. Colors separate all-non-sepsis, mixed, and all-sepsis trajectories.



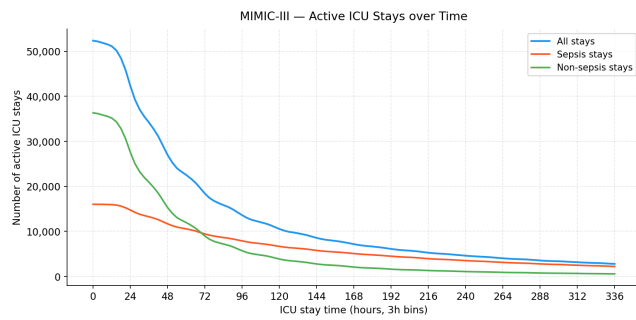
Part 5: Graphical Summaries across ICU Stay

All figures use 3-hour time bins. Source script: [mimic_analysis/plot_icu_curves.py](#).

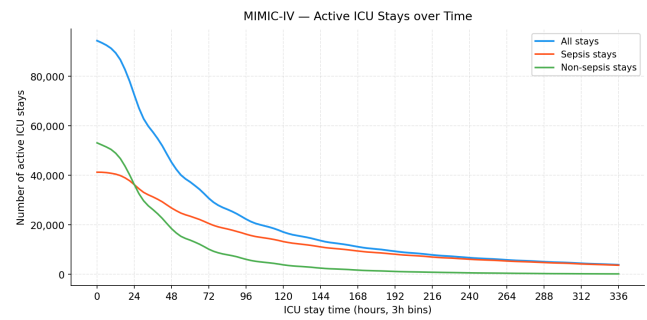
5.1 Active ICU Stays over Time

Each curve shows the number of stays still ongoing at a given ICU hour. Non-sepsis stays discharge earlier, while sepsis stays persist longer and gradually make up a larger share of the remaining ICU population.

MIMIC-III



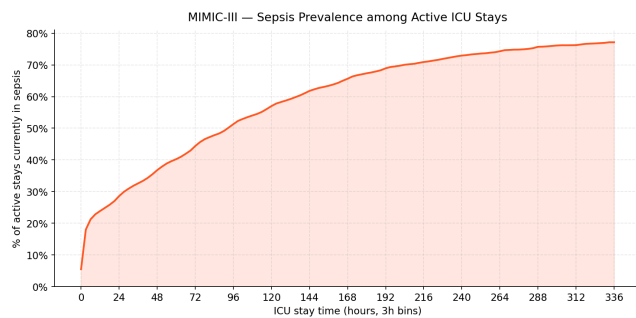
MIMIC-IV



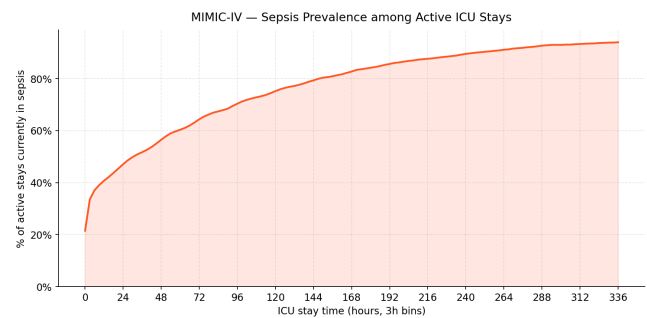
5.2 Sepsis Prevalence among Active Stays

At each hour t , the curve shows the percentage of still-active stays whose sepsis onset has already occurred. The monotonic increase reflects both accumulating onsets and earlier discharge among non-sepsis stays.

MIMIC-III



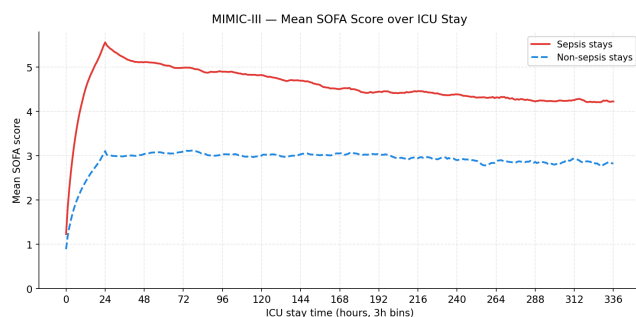
MIMIC-IV



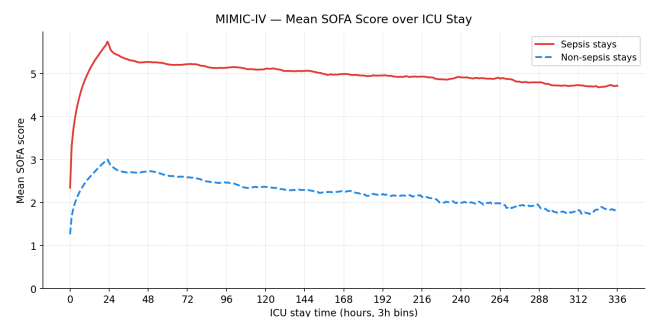
5.3 Mean SOFA Score over ICU Stay

Mean SOFA is computed at each 3-hour bin across all stays still active at that time, stratified by sepsis status. Sepsis stays maintain a consistently higher SOFA trajectory in both datasets.

MIMIC-III



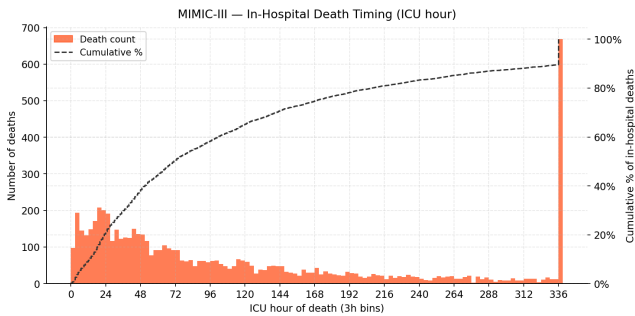
MIMIC-IV



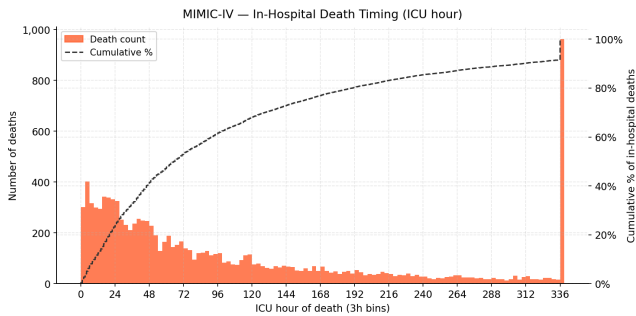
5.4 In-Hospital Death Timing

The histogram shows when in-hospital deaths occur relative to ICU admission, approximated by the last recorded ICU hour for each death stay. The dashed curve shows cumulative deaths over ICU time.

MIMIC-III



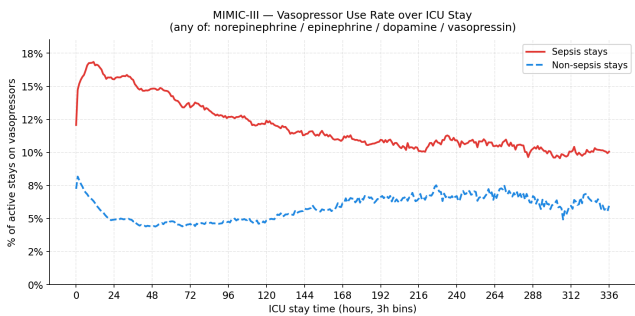
MIMIC-IV



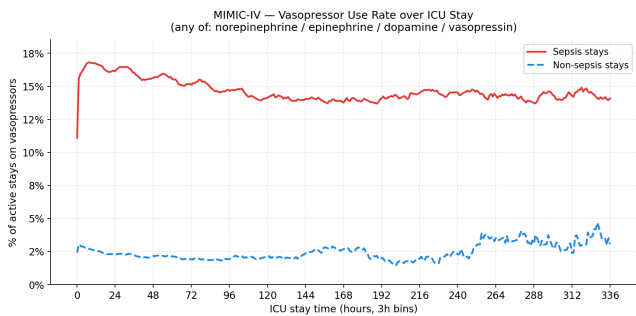
5.5 Vasopressor Use Rate over ICU Stay

The y-axis shows the percentage of currently active stays receiving any vasopressor at each 3-hour bin. Sepsis stays have consistently higher vasopressor exposure in both cohorts.

MIMIC-III



MIMIC-IV



Part 6: ICU Stay Exclusion Flow

Starting from the raw PhysioNet source files, the pipeline applies two sequential filters before building the hourly wide table. Steps with zero exclusions are omitted.

MIMIC-III (source: `ICUSTAYS.csv`, version 1.4)

Item	Remaining stays	Removed	Filter condition
All ICU stays in source database	61,532	—	Full dataset
Exclude paediatric patients	53,362	−8,170	Age < 18 years at ICU admission (predominantly neonates: 8,071 of 8,170)
Exclude admissions without ICU charting	52,894	−468	<code>has_chartevents_data = 0</code> in <code>ADMISSIONS</code> ; no ICU monitoring record exists
Exclude stays without a heart-rate record	52,361	−533	No entry in <code>CHARTEVENTS</code> with itemid 211 (CareVue) or 220045 (MetaVision); time axis cannot be anchored

Item	Remaining stays	Removed	Filter condition
Final wide table	52,361	-9,171 total	—

MIMIC-IV (source: [icu/icustays.csv.gz](#), version 3.1)

Item	Remaining stays	Removed	Filter condition
All ICU stays in source database	94,458	—	Full dataset (MIMIC-IV is adults-only by dataset design; age < 18 = 0)
Exclude stays without a heart-rate record	94,437	-21	No entry in chartevents with itemid 220045; time axis cannot be anchored
Final wide table	94,437	-21 total	—

Notes

- The heart-rate anchor ([intime_hr](#)) defines [hr](#) = 0 for the entire time axis. A stay with no heart-rate record cannot be placed on the hourly grid and is therefore excluded entirely.
- MIMIC-III uses two heart-rate itemids (211 = CareVue, 220045 = MetaVision); MIMIC-IV is MetaVision-only (220045).
- MIMIC-III's [has_chartevents_data](#) flag is a pre-computed column in [ADMISSIONS](#) that marks admissions with at least one chartevent record. Stays failing this check have no ICU monitoring data at all.

Pending Exclusion Decisions

#	Item	Question
1	Pre-ICU rows (hr < 0)	Should all pre-ICU rows be dropped from the wide table entirely, or retained as context features?
2	Stays whose sepsis label depends on pre-ICU data	If pre-ICU rows are removed, any stay where the suspicion window (si_starttime) begins before ICU admission relies on pre-ICU data to set SepsisLabel . Should these stays be excluded, or relabelled using only in-ICU data?
3	Stays with incomplete 24 h SOFA window	The SOFA rolling window is truncated for the first ~24 h of any stay. Should stays shorter than 24 h (or whose only sepsis-eligible window falls within the first 24 h) be excluded to avoid boundary artefacts?
4	Early-onset stays (onset ≤ 24 h or ≤ 48 h)	Per the last meeting, stays with sepsis onset within 24 h (or 48 h) of ICU admission may warrant exclusion. Decision may depend on patient source (ED vs. ward vs. OR transfer) — needs verification of admission source distribution before finalising the threshold.